



nummified skeleton from the Atacama Desert in Chile has been described as "alien." But genetic analysis shows that she was human and may have had a previously unknown bone disorder.

Itacharya S et al. 2018

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Nearly two decades ago, the rumors began: In the Atacama Desert of northern Chile, someone had discovered a tiny mummified alien.

An amateur collector exploring a ghost town was said to have come across a white cloth in a leather pouch. Unwrapping it, he found a six-inch-long skeleton.

Despite its size, the skeleton was remarkably complete. It even had hardened teeth. And yet there were striking anomalies: it had 10 ribs instead of the usual 12, giant eye sockets and a long skull that ended in a point.

Ata, as the remains came to be known, ended up in a private collection, but the rumors continued, fueled in part by a U.F.O. documentary in 2013 that featured the skeleton. On Thursday, a team of scientists presented a very different explanation for Ata — one without aliens, but intriguing in its own way.

Ata's bones contain DNA that not only shows she was human, but that she belonged to the local population. What's more, the researchers [identified in her DNA a group of mutations in genes related to bone development](#).

Some of these mutations might be responsible for the skeleton's bizarre form, causing a hereditary disorder never before documented in humans.

Antonio Salas Ellacuriaga, a geneticist at the University of Santiago de Compostela in Spain who was not involved in the new study, called it "a very beautiful example of how genomics can help to disentangle an anthropological and archaeological dilemma."

"DNA autopsies," as Dr. Ellacuriaga calls them, could help shed light on medical disorders "by looking to the past to understand the present."

The research, published in the journal *Genome Research*, began in 2012, when Garry P. Nolan, an immunologist at Stanford University, got wind of the U.F.O. documentary, "Sirius," while it was still in production.

Dr. Nolan emailed the producers and offered to look for DNA in the mummy. The skeleton's owner agreed to X-ray images as well as bone marrow samples taken from the ribs and right humerus.



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While her elongated head was striking, it wasn't the strangest feature of Ata's skeleton. Despite being the size of a human fetus, about the length of a pen, her bones were as developed in some ways as those of a 6-year-old.

Ralph S. Lachman, an expert on hereditary bone diseases at Stanford University, examined her X-rays. He concluded that her constellation of symptoms did not match any known disease. The scientists reasoned that Ata might have had mutations for a disorder that had never before been described.

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Bhattacharya.

Many of those genes, it turned out, are involved in building skeletons. Some have already been linked to conditions ranging from scoliosis to dwarfism to having an abnormal number of ribs.

But some of Ata's mutations are new to science. It's possible some caused her skeleton to mature quickly even while failing to grow to normal stature.

Ms. Bhattacharya speculates that such a disorder would have caused the child to be stillborn. And she stressed that these mutations are, for now,

Sanchita Bhattacharya, a researcher in Dr. Butte's lab, searched for mutations in Ata's DNA and identified 2.7 million variants throughout the genome. She whittled this list to 54 rare mutations that could potentially shut down the gene in which they were located.

"I was amazed by how much you can tell from the genetic blueprint," said Ms.

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her skeleton to mature quickly even while failing to grow to normal stature.

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Other experts concurred. "There is no single slam-dunk finding that explains the bizarre appearance of this individual," said Daniel G.

MacArthur, a geneticist at the Broad Institute who was not involved in the study.

Yet understanding what happened to Ata might shed light on skeletal deformities seen today. That may require engineering stem cells with each of the 54 mutations, growing them in a dish, and then looking for telling changes in their development.

And Dr. Nolan has heard stories about similar skeletons in other parts of the world. If he were able to examine them, he might discover some of these mutations in their DNA, as well.

Even more direct confirmation might be possible if [researchers paid closer](#)

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Although [there are 24,000 stillbirths in the United States alone](#) each year, doctors generally don't record the features of the fetuses, let alone study their DNA. With so little data, there's no way to know if Ata was unique.

"This could be a trigger to look into more such cases," said Albert Zink, an anthropologist at the European Research Academy in Bolzano, Italy, who was not involved in the new study.

While Dr. Nolan began the project as "a lark," he believes the evidence now requires that the mummy be returned to Chile for proper treatment as human remains.

"One has to respect these things," he said